Master's thesis · Progress Update 2 · weeks 3–4

The best spindle detector depends on the dataset, and retraining learned models only catches up with a good rule

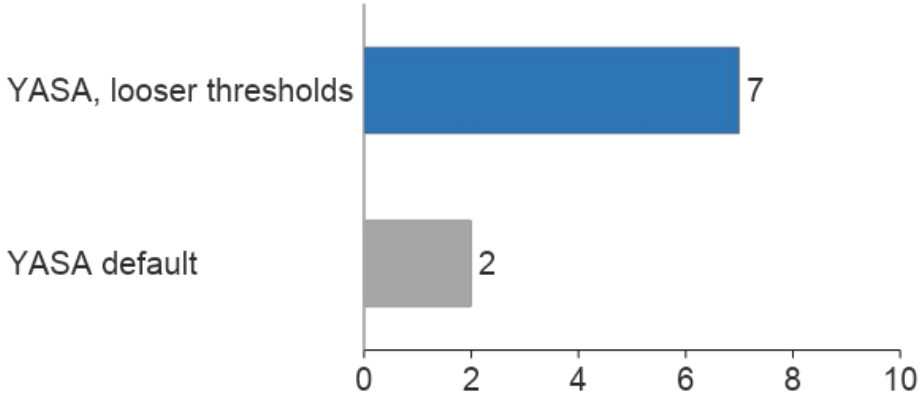
Automatic Detection of Sleep Graphoelements: Benchmarking on Public Datasets and Transfer to in-house Custom Dataset

Nayan Sharma · MAI, THWS

Supervisor: Prof. Dr. Magda Gregorová · 27 October 2026

Update 1 showed that YASA's default settings miss most spindles in one DREAMS window

Expert spindles found in one 60-s window



Plan we left with

- YASA on all of DREAMS
- Proper scoring (F1)
- Open and inspect NAP EEG
- Same pipeline on NAP EEG

Benchmark, validity, transfer, adaptation

Benchmark	5 spindle detectors on DREAMS, NAP EEG and BOAS (4)
Validity	Validate pipeline with published detection numbers
Transfer	Does the DREAMS ranking hold on new data?
Adaptation	What is the cheapest way to adapt a detector?

One pipeline runs 5 detectors and scores them all the same way

1 · Detect

Each detector with its authors' settings

2 · Match

Detection ↔ expert mark, within 0.5 s

3 · Score

Hits, misses, false alarms → F1

- **F1:** did it find the expert's spindles, without false alarms? It is high only when both are true.
- **Scale:** 1 = perfect, 0 = nothing right.

- **N2:** light sleep, where most spindles occur. Main results are scored in N2.

The three datasets

	DREAMS	NAP EEG	BOAS
Recordings	8 × 30 min, patients	30 naps, 17 healthy adults	20 nights, 20 healthy adults
Experts	2	1 per nap	none for spindles
Spindles	355 (expert 1)	2,181	not labelled
Caveat	Expert 1 stops at ~1000 s	One click per spindle, mid-spindle, no duration	11 of 20 nights artefact-heavy
Fix	Score first 1000 s only	Click vs detection centre, ±0.5 s	Agreement and plausibility test

DREAMS: Devuyst et al. (2005). NAP EEG: Mei et al. (2018). BOAS: López-Larraz et al., OpenNeuro ds005555.

Preprocessing: a minimal shared step, then each detector's own

1 · Shared step, per dataset

	DREAMS	NAP EEG	BOAS
Channel	C3 or Cz	C3 (of 61)	C3 (of 6)
Reference	one ear (A1)	linked mastoids	left mastoid
Filter	none	0.1–45 Hz	none
Rate	50–200 Hz	200 Hz	256 Hz

2 · Each detector's own, as published

Detector	Filter, rate	Scaling / threshold
SUMOv2	0.3–30 Hz, 100 Hz	median/IQR, whole night
SUMO	0.3–30 Hz, 100 Hz	z-score, whole night
OpenSpindleNet	none, 250 Hz	z-score per 30 s + wavelet
Mölle 2011	12–15 Hz, 200 ms RMS	1.5 SD, whole night
YASA tuned	1–30 + 12–15 Hz	relative power, corr., RMS

The five detectors

Detector	Type	How it works	Trained on
SUMOv2	learned	U-Net marks every sample as spindle or not	MODA expert labels (MASS)
SUMO	learned	Earlier U-Net, same idea	MODA expert labels (MASS)
OpenSpindleNet	learned	Predicts spindle centre and length per second	DREAMS (so in-sample there)
Mölle 2011	rule-based	12–15 Hz power above 1.5 SD for 0.5–3 s	nothing: a fixed rule
YASA tuned	rule-based	Three rules: sigma power, correlation, RMS	thresholds tuned on DREAMS

SUMOv2: Grieger et al. (2026). SUMO: Kaulen et al. (2022). OpenSpindleNet: Sejak et al. (2025). Mölle et al. (2011), run in wonambi's implementation. YASA: Vallat & Walker (2021). Shown: 5 of 24 spindle detector settings (all in Appendix A).

The pipeline reproduces two published results exactly

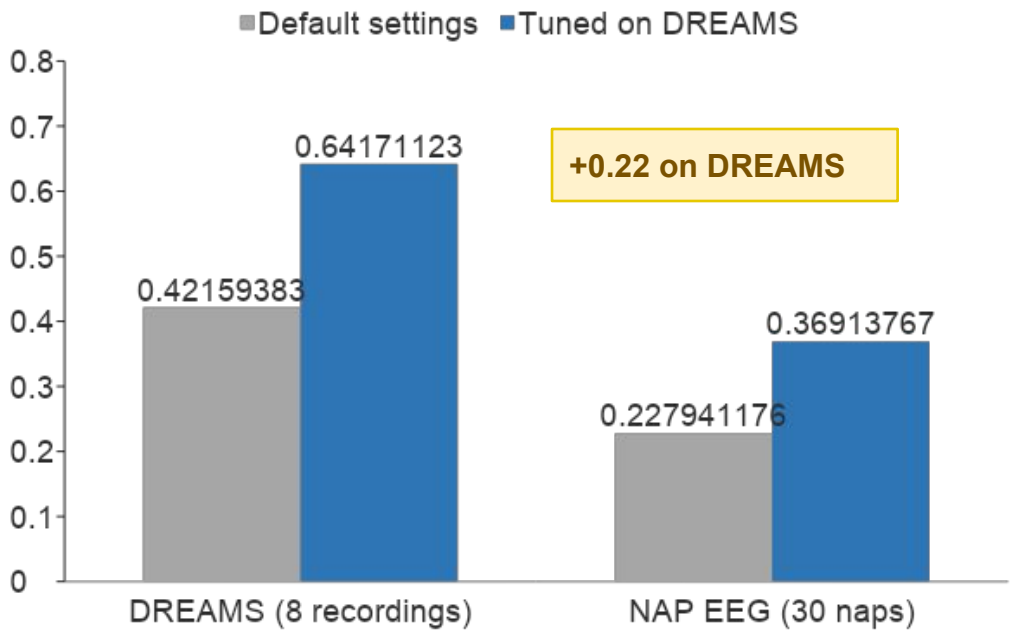
Published result	Paper	Ours
DREAMS detector: sensitivity	70.20%	70.20%
DREAMS detector: false positives	26.44%	26.44%
SUMOv2: F1 vs expert 1	0.53	0.532
SUMOv2: F1 vs expert 2	0.76	0.764

What to take away

- Exact match on both
- Evaluation code is correct

Devuyst et al. (2011); Grieger et al. (2026). Both on DREAMS. SUMOv2 was trained only on MODA (MASS recordings),

On all recordings, tuning YASA's thresholds lifts DREAMS F1 from 0.42 to 0.64

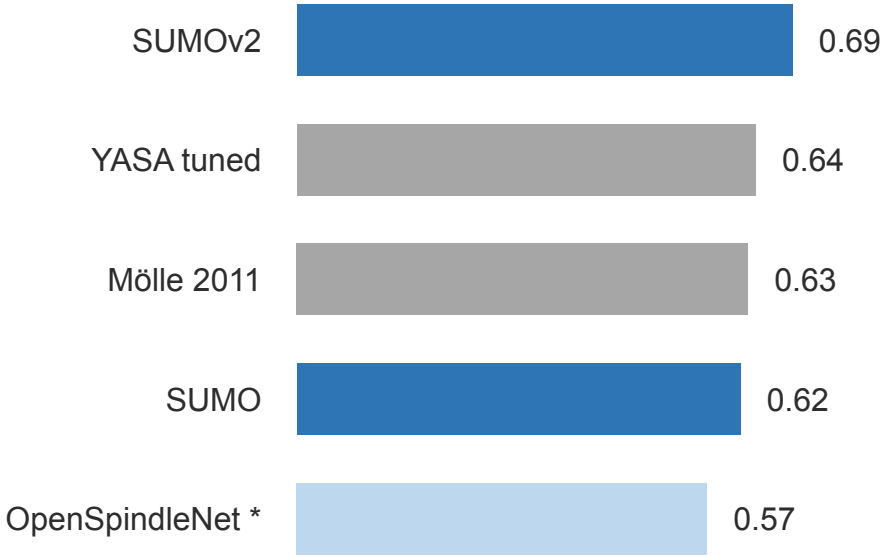


What to take away

- Default finds 27% of DREAMS spindles
- Tuned finds 60%
- Same thresholds on NAP: 0.23 → 0.37

YASA: Vallat & Walker (2021). DREAMS: each recording uses thresholds tuned on the other 7. NAP: thresholds tuned on all 8 DREAMS recordings. N2.

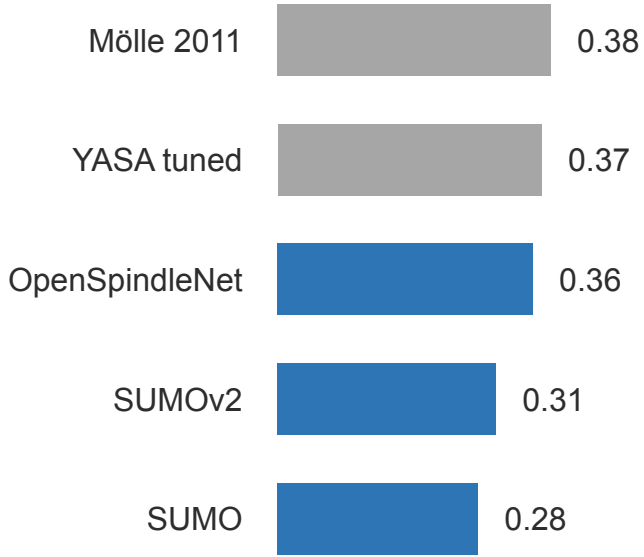
On DREAMS, the best learned model leads, but rule-based detectors are close



What to take away

- SUMOv2 first: 0.69
- Rule-based 0.05–0.06 behind
- 8 recordings: no gap is significant

On NAP EEG every detector scores lower, and rule-based methods lead



What to take away

- Mölle 2011 first: 0.38
- Tuned YASA close: 0.37
- First place not clear in N2
- Labels are single clicks: at ±1.0 s the best rises to 0.52

Spindle density: spindles per minute of N2 sleep

1 · Count

Spindles the detector finds in N2

2 · Divide

by the minutes of N2 sleep

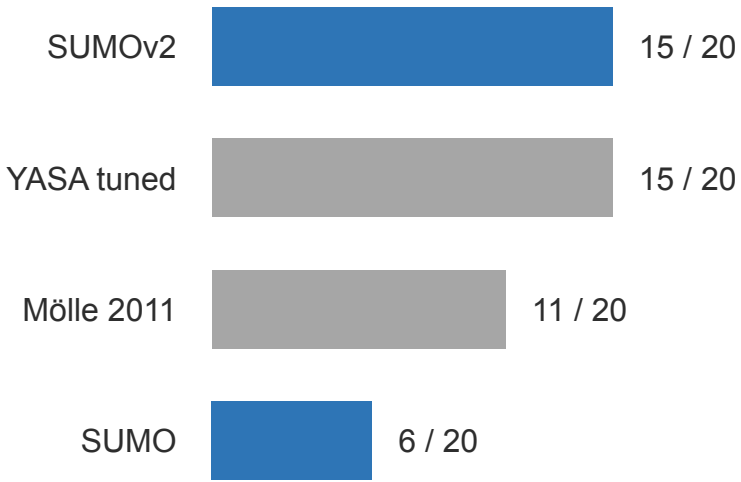
3 · Compare

Healthy adults: about 2–7 per min

- **What:** one of the most commonly reported spindle measures in sleep research.
- **Range:** healthy young and middle-aged adults show roughly 2–7 per minute of N2 on a central channel like C3. The exact range depends on the detector and the channel.
- **On BOAS:** below 2 per minute (usually near 0) means the detector found almost nothing, so we call that night implausible. Older sleepers can be lower.

BOAS: on full nights, artefacts decide which detectors keep working

Nights with a plausible N2 spindle density (≥ 2 per min), of 20



What to take away

- No expert labels: we check if the spindle rate is plausible
- 11 of 20 nights have heavy artefacts

How SUMO was retrained: always tested on people it never saw

	Group 1 4 people	Group 2 4 people	Group 3 3 people	Group 4 3 people	Group 5 3 people
Round 1	**Test**	Train	Train	Train	Train
Round 2	Train	**Test**	Train	Train	Train
Round 3	Train	Train	**Test**	Train	Train
Round 4	Train	Train	Train	**Test**	Train
Round 5	Train	Train	Train	Train	**Test**

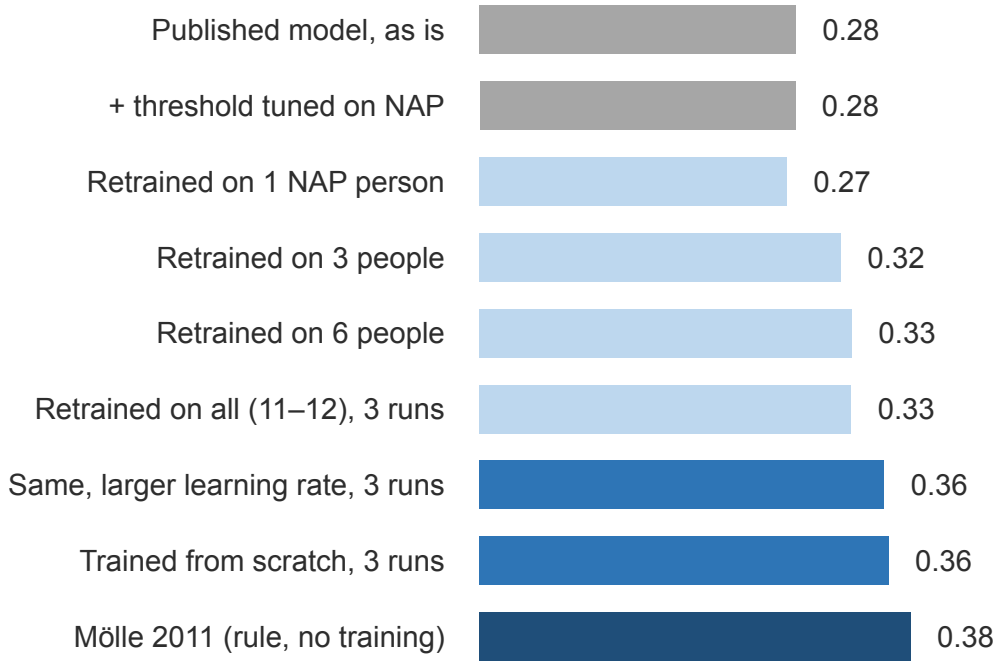
Train = 11–12 people to learn from + 2 for validation (when to stop, which threshold).

Every nap is tested once, by a model that never saw that person. The 5 rounds are then scored together.

How it was trained

- Each expert click → a 0.8 s target
- Learns from 60 s pieces of N2
- Stops when validation stops improving
- 3 runs = trained 3 times, averaged

Retraining SUMO on NAP catches up with, but does not beat, the best rule-based detector

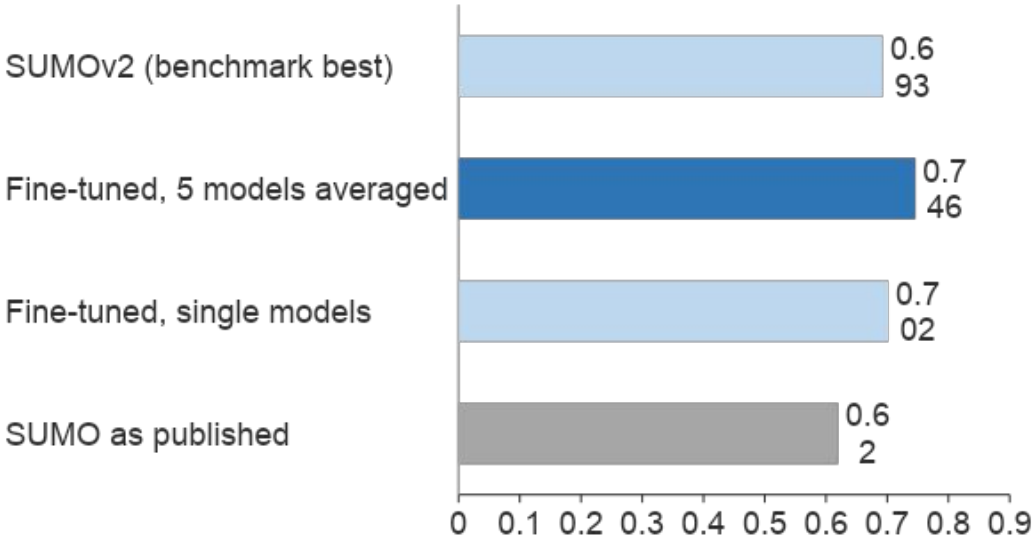


What to take away

- Threshold: no effect
- Retraining on 3+ people: about +0.05
- Same learning rate: retrained = from scratch
- Best ≈ Mölle 0.38

SUMO (Kaulen et al., 2022); Mölle et al. (2011), wonambi.

Training on NAP did not make SUMO worse on DREAMS; it scored higher



What to take away

- No forgetting of old data
- Lead over SUMOv2 not clear

SUMO (Kaulen et al., 2022); SUMOv2 (Grieger et al., 2026).

Conclusions

1. The pipeline is validated: two published results reproduced exactly.

2. No detector is best everywhere: SUMOv2 leads DREAMS, Mölle 2011 leads NAP; neither lead is clear.

3. Thresholds matter: YASA tuned on DREAMS also lifts NAP from 0.23 to 0.37.

4. On full nights, artefacts decide: SUMOv2 and YASA stay plausible on 15 of 20 BOAS nights.

5. Retraining catches up, not past: SUMO reaches 0.36 on NAP vs Mölle 0.38, and does not forget DREAMS.

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Full spindle ranking, N2: DREAMS (vs expert 1) and NAP EEG

Detector	Type	DREAMS	NAP
SUMOv2	Learned	0.69	0.31
A7	Rule	0.67	0.32
A7 (wonambi port)	Rule	0.66	0.31
DREAMS detector *	Rule	0.66	–
A7 in-context	Rule	0.65	0.31
YASA tuned	Rule	0.64	0.37
Mölle 2011	Rule	0.63	0.38
YASA loosened	Rule	0.62	0.29
SUMO	Learned	0.62	0.28
Mölle 2011 (paper recipe)	Rule	0.60	0.34
FASST	Rule	0.59	0.32
Concordia	Rule	0.58	0.34

Detector	Type	DREAMS	NAP
OpenSpindleNet *	Learned	0.57	0.36
Ray 2015	Rule	0.56	0.31
UCSD	Rule	0.56	0.35
Wamsley 2012	Rule	0.55	0.32
FASST2	Rule	0.50	0.30
SSSM v2	Learned	0.47	0.26
Nir 2011	Rule	0.45	0.28
YASA default	Rule	0.42	0.23
Martin 2013	Rule	0.39	0.27
SSSM	Learned	0.30	0.26
Ferrarelli 2007	Rule	0.27	0.16
SSSM strict	Learned	0.02	0.02

Sorted by DREAMS F1. * built or trained on DREAMS labels (not ranked there). wonambi methods = wonambi's implementations. DREAMS onset ±0.5 s (YASA tuned: leave one recording out); NAP centre ±0.5 s, C3. Rule = hand-set filters and thresholds; Learned = neural network trained on labelled EEG.